Data Understanding Report

Week 1 – Data understanding and Initial Analysis

AI Powered Data Analysis Remote Internship

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Dataset: Student Participation Dataset

1.Data Understanding

1. Dataset Purpose

The dataset provided by the Rochester Institute of Technology (RIT) contains historical records of learners and their involvement in educational opportunities offered by RIT. The purpose of the dataset is to organize and analyze learner applications and participation patterns and trends across different opportunity categories.

2. Types of opportunities

The Opportunity Name variable identifies the educational opportunity selected by each learner. The opportunities are categorized as courses, internships, events, engagements and competitions. Analyzing these categories can help identify participation trends and determine which opportunity category had the highest or lowest number of learners. The duration of the opportunity varies by category. Courses generally lasted between one and two years from the start date, while internships lasted approximately one year.

3. How student participation and engagement are recorded

The Status Description variable records the outcome of a learner's application or participation, such as started, allocated to a team, waitlisted, withdrew, rejected, or dropped out. Analyzing this variable can identify participation patterns, such as whether more learners started an opportunity than withdrew.

4. Number of records

The dataset originally contains 8,559 populated rows, including the header row. After removing 5 rows during the data-cleaning process, 8,554 rows remain. Excluding the header row, the cleaned dataset contains 8,553 learner records.

5. Number of variables

There are 16 variable columns, spanning from column A to column P.

6. Date range covered

The dataset covers the period from 2022 to 2024, with the application period ranging from 5 October 2022 to 11 March 2024. This represents the period during which the 8558 learner application records were recorded.

Data Variable	Data Range Covered
Apply Date	5 October 2022 – 10 April 2024
Learner Signup Date Time	5 January 2023 – 2 April 2024
Opportunity Start Date	3 November 2022 – 31 May 2024
Opportunity End Date	1 March 2023 – 24 December 2025
Entry created at Date	11 March 2024

In conclusion, the dataset offers a structured historical perspective on student participation, application activity, and engagement, which can be utilised to identify trends and patterns in student involvement.

2.Data Dictionary

We have 8,559 Rows and 16 Columns in the original dataset.

	Variable	Description	Entry Data type	Ideal Data type	Analytical relevance
1	**Learner Signup DateTime**	The date and time the learner registered on the platform.	Number	Datetime (MM/DD/YYYY)	Useful for signup/participation trends over time.
2	**Opportunity Id**	Unique identification number for each opportunity.	Text	Categorical Text	Key for grouping and joining. Use it instead of the name.
3	**Opportunity Name**	Title or specific name of the program.	Text	Categorical Text	Shows the distinct names of each program that was offered.
4	**Opportunity Category**	The class of the program.	Text	Categorical	Helps in grouping and summary reports.
5	**Opportunity End Date**	Closing date of the opportunity.	Number	Datetime	Useful for opportunity duration and timing analysis.
6	**First Name**	Learner's first name	Text	Text	Helps identify duplicate entries.
7	**Date of Birth**	Learner's date of birth	Number	Date (MM/DD/YYYY)	Demographic profiling and age-group comparisons**.**
8	**Gender**	Learner's sex.	Text	Categorical Text	Used to check participation and outcome differences by gender.
9	**Country**	Learner's country	Text	Categorical Text	Geographic analysis.
10	**Institution Name**	Learner's school or university	Text	Text	For grouping and assessing success rates.
11	**Current/Intended Major**	Learner's current or intended course of study.	Text	Text	Can show participation patterns by academic field
12	**Entry created at**	When the record was created in the system.	Number	Datetime	May be needed for data retrieval or other general reports.
13	**Status Description**	The status of the signed-up learner's application.	Text	Categorical	Important for understanding outcomes/participation stages.
1 4	**Status Code**	The unique code of the status in the previous column	number	number	It maps one-to-one to Status Description, so it is for modeling and treats the code as a label.

15	**Apply Date**	The date and time the learner applied for the program.	number	Datetime	Used for application trends and seasonality, and to compute time from sign-up to application.
16	**Opportunity Start Date**	The date and time the opportunity started.	number	Datetime	Used for timing analysis and duration.

Relevant Notes

- Identifying the correct data type used in entering the variables was done, and the ideal data type was given for proper data structuring.
- Variables like entry date may have little or no significance to the analysis at hand but may be used for future retrieval of information about the historical student data.

3. Data Quality Assessment

The dataset was assessed for missing values, duplicate records, inconsistencies, and data-type issues.

Missing Values

- Opportunity Start Date has significant missing values, with 3,794 missing records (44.33%).
- Minor missing values were also observed in fields such as Institution and Major.
- The high missing number of values in Opportunity Start Date may limit analyses involving opportunity timelines.

Duplicate Records

- No exact duplicate records were identified in the dataset.

Inconsistencies

- No major internal inconsistencies were identified in the main categorical variables.

Data Types

- The following date/time fields are currently stored as object rather than datetime:
    - Learner SignUp DateTime
    - Opportunity End Date
    - Entry created at
    - Apply Date
    - Opportunity Start Date
- These fields should be converted to appropriate datetime types during cleaning to support reliable analysis and date-based calculations.

Overall Assessment

The dataset is usable for exploratory analysis, but some processing is required, particularly for handling missing values and converting date/time fields. The identified issues should be addressed during the data-cleaning stage.

4.Data Cleaning

Overview

The dataset was reviewed and cleaned before analysis to improve consistency, remove obvious test records, handle missing values, standardize categorical information, and correct identifiable formatting issues.
The original dataset contained 8,559 rows and 16 columns. After removing five clearly identifiable test/junk records, the working dataset contains 8,554 rows and 16 columns.
The cleaning process focused on correcting issues that could be identified with reasonable confidence while avoiding changes that could introduce assumptions or alter potentially valid records.

Duplicate Records

Exact duplicate rows were checked using the complete dataset.

- Exact duplicate rows found**:** 0
- No records were removed for exact duplication.

Repeated learner-opportunity combinations were also investigated. Some learners appeared more than once for the same opportunity, sometimes with different status descriptions. These records were retained because the repeated entries could represent different application or status events and could not safely be classified as duplicates.

Missing Values

Missing values were examined across all columns.
The following missing values were addressed:

Institution Name

A small number of records had missing institution names.

- Missing values were replaced with “Not Provided”

Some clearly invalid institution entries containing email addresses were also replaced with Not Provided.

Current/Intended Major

A small number of records had missing major information.

- Missing values were replaced with “Not Provided”

Opportunity Start Date

A significant number of records did not contain an Opportunity Start Date.

- These values were not filled or estimated**.**
- They were retained as missing because there was insufficient information to determine the correct dates.

This avoids adding fabricated dates into the dataset.

Date and Time Cleaning

The date-related columns were examined for inconsistent formats and abnormal time values.
The following columns were processed:

- Learner SignUp DateTime
- Apply Date
- Opportunity Start Date
- Opportunity End Date
- Entry created at

The cleaning process handled identifiable Excel serial-date values and attempted to normalize malformed time representations.
The other date fields were converted into proper datetime values where possible.

Learner SignUp DateTime Limitation

The Learner SignUp DateTime column contained several problematic values with unusually large hour components, for example values similar to:
06/10/2023 58:20

Some of these values could not be reliably corrected because the intended time/date interpretation was ambiguous.
These records were therefore not forcibly converted or deleted**.**

The column was retained in its original textual form where necessary to prevent valid dates from being incorrectly converted to missing values.

Country Standardization

Country names were reviewed for inconsistent naming.
One identified variation:
Korea, Republic of South Korea
was standardized to South Korea.
Other country values were retained where there was no sufficiently clear reason to change them.
The objective was to improve consistency without making assumptions about uncommon or unusual country names.

Institution Name Standardization

Institution names were cleaned for basic consistency.
The following operations were performed:

- Leading and trailing whitespace was removed.
- Capitalization was standardized using title case.
- Clearly identified variations of the same institution were combined where appropriate.

For example:

- Saint Louis
- St Louis University

were standardized to “Saint Louis University”.
Entries that appeared unusual but could not confidently be identified as incorrect were retained.

First Name Standardization

The First Name column contained inconsistent capitalization.
Examples included names written in lowercase, uppercase, and mixed case.
Names were standardized using title case and leading/trailing whitespace was removed.
This was done to improve consistency for grouping, filtering, and analysis.

Current/Intended Major Standardization

The Current/Intended Major column was standardized by:

- Removing leading/trailing whitespace.
- Applying consistent title-case formatting.
- Replacing missing values with Not Provided.

No attempt was made to merge every semantically similar major because doing so could require subjective interpretation.

Test and Junk Records

The dataset contained several obvious test/junk records.
Five records were identified as clear test data and removed.
These included records with values such as:

- Test
- test
- asdads

These records were sufficiently obvious to be considered data-entry/testing artifacts rather than meaningful learner records.

Other Test-Like Records

Some additional records appeared unusual or potentially related to testing, including records containing values such as:

- Student
- Potencia Student in SLU

These records were not removed because they contained other potentially meaningful information and could not be confidently classified as invalid.

Learner Records with Multiple Entries

Some learners appeared multiple times in the dataset for the same opportunity.
These records were investigated using combinations of:

- First Name
- Date of Birth
- Opportunity ID
- Status Description

Some repeated combinations had different status descriptions.

Because these could represent legitimate application/status activity rather than duplicate records, they were retained.

Date of Birth Review

Some records contained relatively young dates of birth that could indicate potentially young participants.
These records were reviewed but were not deleted**.**
The presence of an unusual age alone was not considered sufficient evidence that the record was invalid. Such records were therefore retained for analysis.

Data Types and Formatting

The dataset was reviewed for inconsistent data types, particularly in date-related columns.
Excel serial date values were identified and converted where applicable.
The main date fields, excluding problematic signup values that could not be safely interpreted, were converted into appropriate datetime representations for analysis.
Text fields were stripped of unnecessary leading/trailing whitespace where appropriate.

Final Dataset

After cleaning, the dataset contains:

- 8,554 rows
- 16 columns

The resulting dataset is considered suitable for analysis while retaining a small number of documented data-quality limitations.
The remaining issues were intentionally left unresolved where there was insufficient information to determine the correct value with confidence.

5.Historical Analysis

Application Trends Over Time

The Learner SignUp DateTime column was originally stored as an object/string field and contained some malformed date values. A temporary Signup_Date column was therefore created using date conversion with invalid values converted to missing (NaT). This allowed reliable dates to be used for historical analysis without modifying the original cleaned data.

Monthly signup records were then grouped by month to identify changes in participation over time. The monthly results showed variation rather than a consistent upward or downward trend. The highest monthly count occurred in January 2024 with 1,547 records, while August 2023 also showed high activity with 1,133 records**.**
Because monthly figures can fluctuate significantly, the records were also grouped into quarters:

Quarter	Records
2023 Q1	500
2023 Q2	1,435
2023 Q3	2,503
2023 Q4	1,270
2024 Q1	2,739
2024 Q2	11*

*Q2 2024 is incomplete because the dataset only goes into early April 2024 and therefore should not be interpreted as a full-quarter decline.

The quarterly results indicate that participation increased greatly from Q1 to Q3 2023, declined in Q4 2023, and increased again in Q1 2024. Overall, the dataset shows fluctuating participation pattern rather than a consistent long-term growth or decline pattern**.**

Growth and Decline Patterns

Month-over-month percentage change was calculated to identify periods of particularly strong increases or decreases.
The analysis showed several substantial changes, including:

- A strong increase from March to June 2023.
- A major increase in August 2023.
- Consecutive monthly declines from September through November 2023.
- A strong rebound in December 2023 and January 2024.
- A decline after January 2024.

The large decline observed in March and April 2024 should be interpreted with caution because the dataset may not contain a complete period for these months, particularly April.

Significant Participation Observations

The Status Description field was analyzed using record counts and percentages to understand the distribution of participation outcomes.

Status	Count	Percentage
Rejected	3,568	41.72%
Team Allocated	3,276	38.30%
Started	764	8.93%
Dropped Out	616	7.20%
Waitlisted	109	1.27%
Applied	105	1.23%
Withdraw	86	1.01%
Rewards Award	29	0.34%

The most significant observation is that Rejected and Team Allocated were the two dominant status categories**.**

Records classified as Started represented 8.93%**,** while Dropped Out represented 7.20%**.** The remaining statuses each represented a relatively small proportion of the dataset.
These percentages describe the distribution of records by status and should not be interpreted as unique-learner rates because repeated learner/opportunity records are present in the dataset.

Opportunity Duration Statistics

Opportunity duration was calculated as:

Opportunity End Date − Opportunity Start Date
Initially, 4,760 records had both dates available. However, 600 records produced negative durations**,** indicating inconsistent date values where the recorded end date occurred before the start date. These records were excluded from the duration statistics.

The resulting statistics for 4,160 valid records were:

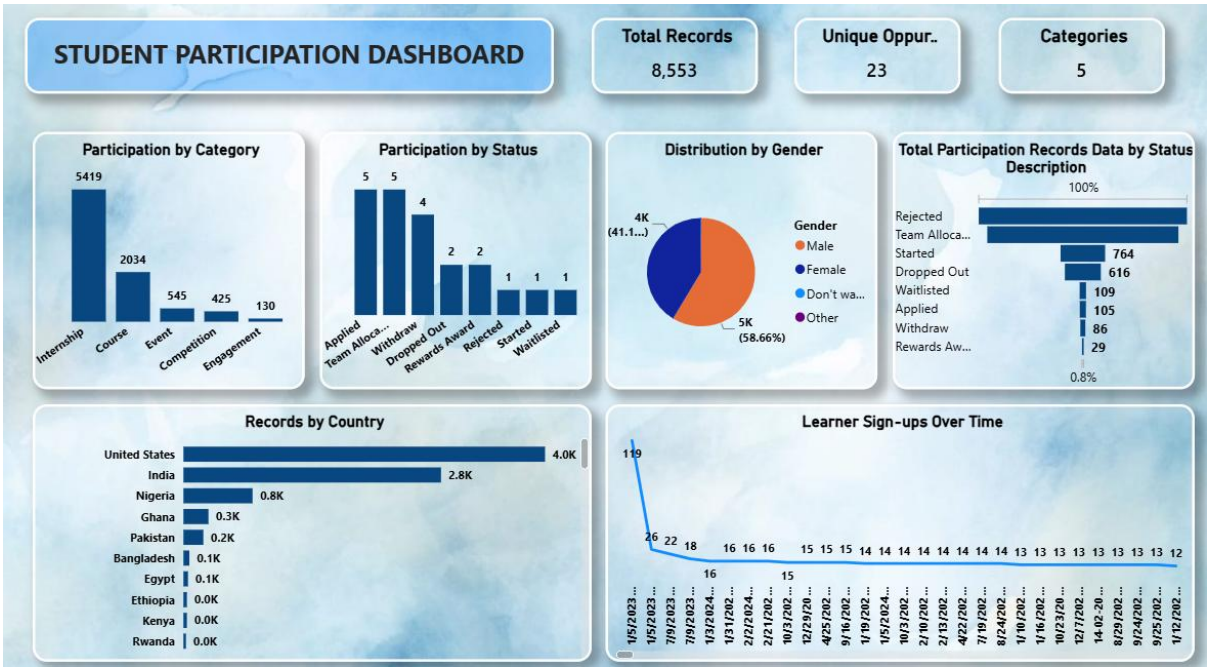
Statistic	Value	Analytics extracted
Count	4,160	Valid opportunity durations
Mean	328.3 days	Average opportunity lasted ~328 days
Median	94 days	Middle opportunity lasted 94 days
Minimum	6 days	Shortest valid opportunity
Q1 (25%)	63 days	25% lasted 63 days or less
Q3 (75%)	604 days	75% lasted 604 days or less
Maximum	913 days	Longest opportunity

The mean being higher than the median indicates variation in opportunity duration with some long-duration opportunities increasing the average.
This analysis represents opportunity duration rather than individual learner engagement duration**,** since the dataset does not contain a reliable individual participation start and end period.

Limitations

- 95 signup date values could not be reliably converted and were excluded from time-based trend calculations.
- Q2 2024 is incomplete and therefore cannot be compared directly with complete quarters.
- Some March/April 2024 records may also represent incomplete period coverage.
- 600 records contained negative opportunity durations and were excluded from duration statistics.
- Status percentages represent records rather than unique learners.
- Opportunity duration should not be interpreted as individual learner engagement duration.

6.Visualisations



The PowerBI dashboard provides an overview of learner participation across 8,558 records, covering 5 opportunity categories and 23 unique opportunities. It summarizes participation by opportunity category, status, gender, country, and opportunity category by status. The dashboard also presents learner signup activity over time to provide a historical view of participation patterns.

The visualizations highlight several key observations: Internships represent the largest opportunity category, supporting our categorical distribution analysis, followed by Courses, while Rejected and Team Allocated account for the largest shares of participation statuses. The geographic distribution shows participation from multiple countries, with the United States and India representing the largest groups in the displayed data. The time-series visualization shows that learner signup activity varies considerably across the observed period rather than following a consistent trend.

Note: The dashboard is intended as a visual summary of the dataset. The underlying analysis was used to validate the displayed counts and identify relevant trends and data-quality limitations.

Team Charter:

Team Charter.docx

Use of AI

AI Tool Used

- ChatGPT was used as an analytical support tool during the data understanding and historical analysis process.

Prompts / Tasks Used
AI assistance was used to:

- Explain appropriate pandas functions/operations for monthly and quarterly grouping.
- Assist in Interpreting/documenting month-over-month changes and participation statistics.
- Help with fixing potential data-quality issues in date-based calculations.
- Help structure/document the results.

Insights Generated
AI-assisted analysis helped identify:

- Data-quality issues such as invalid signup dates, incomplete periods, and negative opportunity durations.
- key patterns driven from outputs of raw data.

Insight Validation Process

AI-generated suggestions and interpretations were validated against the actual dataset outputs. Python/pandas calculations were run on the cleaned dataset, and the resulting counts, percentages, dates, and duration statistics were reviewed before being included in the analysis.

AI was therefore used as a support and interpretation tool, while the actual calculations and final conclusions were based on the dataset and verified outputs.

ChatGPT was also used to assist with sentence structure clarity, and the use of technical data-analysis terms.
Prompts included:
“Restructure sentences to explain the data concepts concisely and professionally?”
“Explain the meaning of the dataset’s data range."